IRIS INTELLIGENCE

Explore the Data. Understand the Model. Predict the Species.

An interactive Iris flower classification and machine learning analytics platform

96.67%	0.9666	29/30
Held-out accuracy	Held-out macro F1	Correct predictions

Abstract and objective

Five classical classifiers were trained and tuned to predict one of three Iris species from four measurements in centimeters. Decision Tree was selected using training-only cross-validated macro F1. The independent holdout yielded 96.67% accuracy. This is evidence for a specific small benchmark split, not a claim of universal botanical reliability.

The authentic dataset

Source	Kaggle / saurabh00007/iriscsv
Observed sample	150 flowers; 50 per species
Inputs	Sepal length, sepal width, petal length, petal width (cm)
Target	Iris-setosa, Iris-versicolor, Iris-virginica
Integrity	No missing values; 3 repeated measurement rows retained; Id excluded

Complete workflow

Acquire and validate the CSV; explore measurements; group identical observations; reserve the holdout; tune five pipelines within training CV; select one model; evaluate once; save the pipeline and reports; support interactive numerical inference.

Acquisition: direct HTTPS download from official Kaggle dataset API. The source checksum, exact row indices and package versions are included in the machine-readable experiment metadata.

Methodology and experimental setup

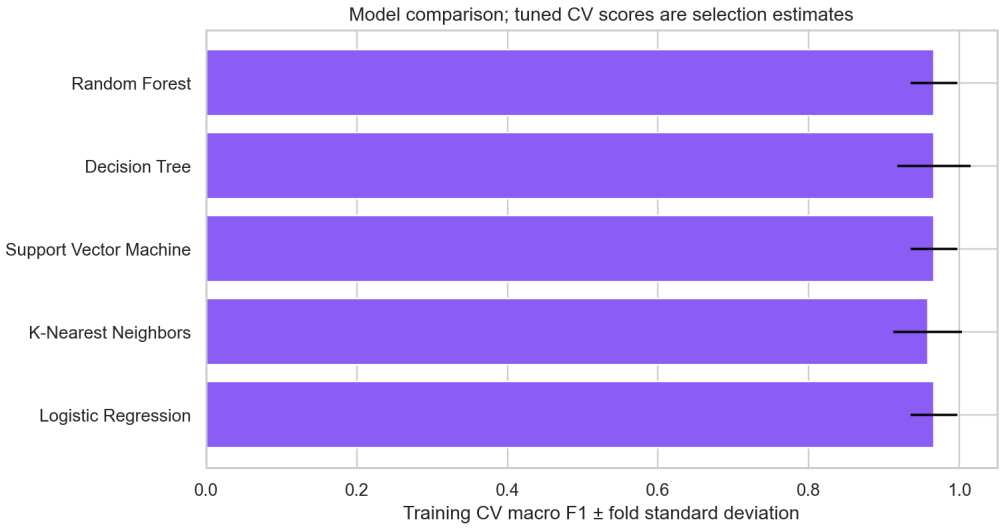
Seed 42. An 80/20 stratified split of distinct four-feature measurement groups produced 120 training rows and 30 test rows. Identical measurements cannot cross boundaries. Five-fold shuffled StratifiedGroupKFold is used inside GridSearchCV. All observations remain; no imputation or arbitrary outlier removal occurs.

Logistic regression, KNN and SVM use StandardScaler inside their pipelines. Trees use passthrough. Scaling and tuning see training data only. The saved pipeline is fitted on the training partition and is never refitted on test observations.

Algorithm	Hyperparameters explored
Logistic regression	C: 0.1, 1, 10
KNN	k: 3, 5, 7, 9; uniform/distance; p: 1, 2
SVM	Linear/RBF; C: 0.1, 1, 10; RBF gamma: scale, 0.1
Decision tree	Depth: 2, 3, unlimited; min leaf: 1, 3
Random forest	Trees: 100, 200; depth: 3, unlimited; min leaf: 1, 2

Actual cross-validation comparison

Algorithm	Macro F1	Fold SD	Accuracy
Logistic Regression	0.9663	0.0313	0.9652
K-Nearest Neighbors	0.9579	0.0458	0.9561
Support Vector Machine	0.9663	0.0313	0.9652
Decision Tree	0.9665	0.0488	0.9644
Random Forest	0.9663	0.0313	0.9652



Winner: **Decision Tree**. Best settings: {'model__max_depth': 3, 'model__min_samples_leaf': 3}. Macro F1 is the primary criterion; numerical ties use CV accuracy then a fixed simplicity order. The tiny observed advantage is not evidence of general superiority.

Fold SD describes variation, not a confidence interval. Tuned CV scores are optimistic selection estimates. Only the selected model receives a holdout evaluation.

Independent held-out evaluation

Metric	Macro average	Weighted average
Precision	0.9697	0.9697
Recall	0.9667	0.9667
F1	0.9666	0.9666

Accuracy is the fraction correct. Precision measures correctness among predicted members of a class; recall measures recovery among actual members. F1 balances precision and recall. Macro averages treat species equally; weighted averages account for support.

Species	Precision	Recall	F1	Support
Iris-setosa	1.000	1.000	1.000	10
Iris-versicolor	1.000	0.900	0.947	10
Iris-virginica	0.909	1.000	0.952	10

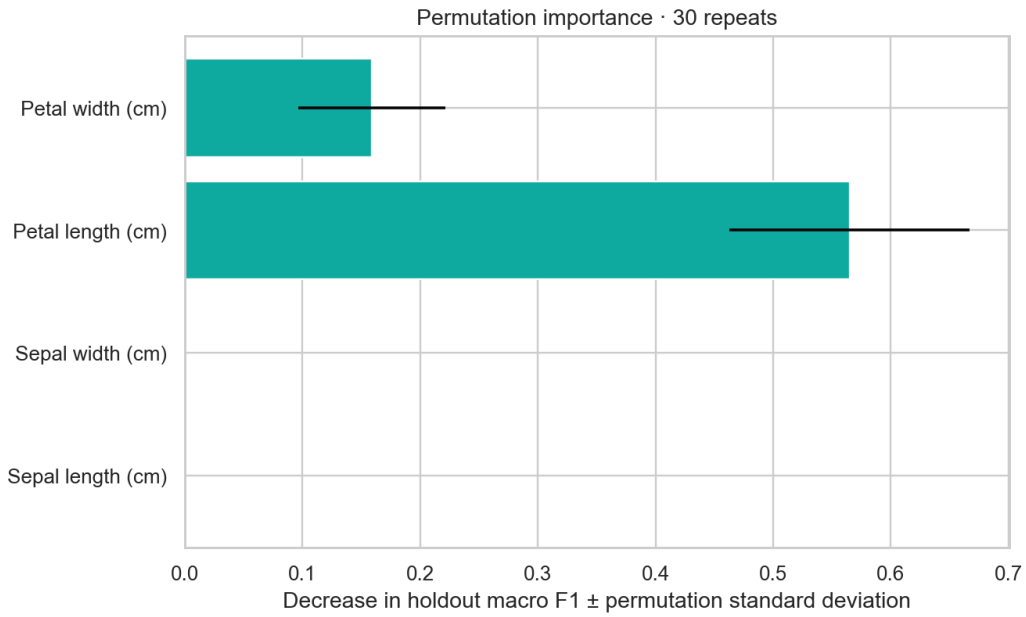


Real error analysis

CSV data row 77 (zero-based): actual **Iris-versicolor**; predicted **Iris-virginica**. Measurements: sepal 6.7 x 3.0 cm; petal 5.0 x 1.7 cm.

Species overlap is a plausible explanation for ambiguity, not a proven cause of this particular error. All held-out examples and normalized confusion counts are available in the dashboard and CSV/JSON outputs.

Feature relevance and scientific discussion



Permutation importance shuffles one held-out feature at a time and measures the decrease in macro F1 over 30 repetitions. Error bars show shuffle variation. Correlated features can substitute for each other; these are global model diagnostics, not causal or per-prediction explanations. Importance was computed after selection and did not inform retuning.

Descriptive EDA shows strong petal-feature separation for setosa and more overlap between versicolor and virginica. The full-dataset PCA is an exploratory map only; it is not used by the predictive pipeline. Similar training examples and what-if curves provide context and sensitivity, not causal evidence.

Limitations and future improvements

The 150-row historical benchmark is small, balanced and restricted to three species. One 30-row holdout has substantial sampling variation. Class probability estimates have not undergone an independent calibration study. Range checks flag extrapolation but cannot detect every unfamiliar feature combination. The classifier receives measurements, never photographs.

Future work should add externally measured flowers, nested cross-validation, independent calibration and measurement-noise analysis. The current system prioritizes honest evaluation, source traceability and understandable local inference.

Conclusion and reproducibility

The project implements the complete classification lifecycle and exposes it through seven local Streamlit workspaces. Reproduce using the pinned Python environment, authentic source CSV, seed 42 and recorded split/search settings. Run **python -m src.train** and **python -m pytest -q**. The source, metadata, executed notebook and Markdown report supply the complete audit trail.

References

Required Kaggle Iris.csv

Scikit-learn leakage prevention

Scikit-learn permutation importance

Streamlit documentation